

August 13. polynomial congruence.



Let K be a field and consider $K[x]$.

It is an $\mathbb{E}D$. In particular, given

polynomials $f(x)$, $g(x) \neq 0$ in $K[x]$.

$\exists!$ $q(x)$, $r(x) \in K[x]$ such that

$$f(x) = q(x)g(x) + r(x) \text{ where,}$$

$$r(x) = 0 \quad \text{or} \quad \deg r(x) < \deg g(x).$$

So we may define the congruence class

as before. That is, for a fixed $g(x) \neq 0$

we define $K[x] \bmod g(x)$

be the collection of equivalence classes

$$[a(x)] = \{ f(x) \in K[x] : f(x) \equiv a(x) \pmod{g(x)} \}$$

Note by the division algorithm.

every $f(x)$ is congruent to a unique

$r(x)$ with $\deg r(x) < \deg g(x)$.

eg. $\mathbb{R}[x] \pmod{x^2+1}$ consists of

(1) $[r]$ for $r \in \mathbb{R}$.

(2) $[ax+b]$ for $a, b \in \mathbb{R}$

and nothing more, because the

remainder modulo x^2+1 has degree ≤ 1 .

In particular $\mathbb{R} \subseteq \mathbb{R}[x]$, and $[x^2] = [-1]$

Note $k[x] \bmod g(x)$ has a ring structure

$$\text{for } [a(x)] + [b(x)] = [(a+b)(x)]$$

$$[a(x)] \cdot [b(x)] = [a(x)b(x)]$$

eg. Note $k[x] \bmod x$ consists of

$[r]$ for $r \in k$ and nothing more.

so $k[x] \bmod x$ is actually a field

One may ask. for which $g(x) \in k[x]$.

is $k[x] \bmod g(x)$ a field.

One possible answer is that

for all $g(x)$ of deg 1.

eg. $\mathbb{R}[x] \bmod x^2+1$ is a field.

Note $[r]$ for $r \in \mathbb{R} - \{0\}$ is clearly

a unit. For $[ax+b]$ where $a \neq 0$.

note that for $c \neq 0$

$$[ax+b][cx+d] = [acx^2 + (ad+bc)x + bd]$$

$$= [(ad+bc)x + (bd-ac)] = [1]$$

iff $ad+bc = 0$, $bd-ac = 1$ in $\mathbb{R}$

$$\text{As } a \cdot c \neq 0, \quad d = -bc/a$$

$$\text{and } -b^2c/a - ac = 1$$

$$\text{implies } -b^2c - a^2c = a$$

$$(a^2 + b^2)c = -a$$

As $a \neq 0$, $a^2 + b^2 \neq 0$

hence $c = -a / (a^2 + b^2)$

So $[ax+b]$ is a unit whenever $a \neq 0$

In particular $\mathbb{R}[x] \bmod x^2+1$ is a field.

One may consider the function

$$\rho: \mathbb{R}[x] \bmod x^2+1 \rightarrow \mathbb{C}$$

$$ax+b \mapsto ai+b$$

and show ρ is a ring isomorphism

and conclude $\mathbb{R}[x] \bmod x^2+1 \cong \mathbb{C}$.

which implies $\mathbb{R}[x] \bmod x^2+1$ is a field.

eg. $\mathbb{R}[x] \bmod x^2$ is not a field

because it is not a domain

$$\text{for } [x] \cdot [x] = [x^2] = [0].$$

Lemma: let K be a field and

$g(x) \in K[x]$. Then $K[x] \bmod g(x)$

is a field iff $g(x)$ is irreducible in $K[x]$.

proof: Note $g(x)$ is irreducible in $K[x]$

iff $(g(x))$ is a maximal ideal in $K[x]$.

Also note that $[f(x)] \neq [0]$

in $K[x] \bmod g(x)$ iff $f(x) \notin (g(x))$.

For the division algorithm says

$$f(x) = q(x)g(x) + r(x)$$

$$\text{for } r(x) = 0 \quad \text{or} \quad \deg r(x) < \deg g(x)$$

In particular. $q(x)g(x) \in (g(x))$, $r(x) \notin (g(x))$

$$\text{and } r(x) = 0 \quad \text{iff} \quad f(x) \in (g(x))$$

$$r(x) \neq 0 \quad \text{iff} \quad f(x) \notin (g(x))$$

So $[f(x)] \neq 0$ in $k[x] \text{ mod } g(x)$

iff $f(x) \notin (g(x))$ which is maximal

$$\text{iff } (f(x), g(x)) = k[x]$$

$$\text{iff } \exists a(x), b(x) \in k[x]$$

such that $a(x)f(x) + b(x)g(x) = 1$

$$\text{iff } [a(x)] [f(x)] = [1]$$

$$\text{in } K[x] \text{ mod } g(x)$$

eg. Consider $\mathbb{R}[x] \xrightarrow{\rho} \mathbb{C}$ sending $x \mapsto i$

This is a ring homomorphism.

The kernel $\ker \rho = (x^2 + 1)$

Since ρ is not the zero map

$$\ker \rho \neq \mathbb{R}[x].$$

But $(x^2 + 1)$ is maximal in $\mathbb{R}[x]$.

and $\ker f$ is an ideal.

$$\text{So } \ker f = (x^2 + 1).$$

This is another derivation that

$$\mathbb{R}[x] \text{ mod } x^2 + 1 \text{ is isomorphic to } \mathbb{C}$$